Companion
to

ChatGPT Teaches TikZ

Chapter 26: Advanced Coordinate Systems and Calculations (A)

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26.1 What This Companion Adds

Chapter 26 replaced measured positions by geometric relationships. This Companion extends that method to projections, extrapolation, signed offsets, orthogonal routing, and constructions whose correctness can be tested by moving the original points.

Design

Treat a derived coordinate as a statement of dependence. Its source should explain why the point is there, not merely where it happened to be in one version of the picture.

26.2 A Coordinate-Calculation Vocabulary

Form or option	Purpose
`$(A)!t!(B)$`	Interpolates or extrapolates along line `AB`.
`$(A)!d!(B)$`	Moves a fixed distance from `A` toward `B`.
`$(A)!d!90:(B)$`	Rotates the direction before moving.
`$(A)!(P)!(B)$`	Projects `P` onto the line through `A` and `B`.
`(A -| B)`	Uses the x-coordinate of `A` and y-coordinate of `B`.
`(A |- B)`	Uses the x-coordinate of `B` and y-coordinate of `A`.
`name path=`	Names a path used in a construction.
`name intersections=`	Computes crossings of two named paths.
`by=`	Gives computed intersections usable names.

26.3 Interpolation and Extrapolation

The familiar expression

```
\coordinate (M) at ($(A)!0.5!(B)$);
```

places `M` halfway from `A` to `B`. The parameter need not lie between zero and one.

```
\coordinate (P) at ($(A)!1.25!(B)$);
\coordinate (Q) at ($(A)!-.25!(B)$);
```

Point `P` lies beyond `B`; point `Q` lies on the opposite side of `A`. This is useful for extending construction lines without measuring their direction.

Common Mistake

A dimensionless value is a fraction of the vector from `A` to `B`. A value with a unit, such as `8mm`, is a fixed distance. Do not interchange the two meanings.

26.4 Signed Fixed-Distance Offsets

```
\coordinate (P) at ($(A)!8mm!(B)$);
\coordinate (Q) at ($(A)!-8mm!(B)$);
```

The first point follows the direction toward `B`; the second follows the reverse direction. Both remain exactly 8 millimeters from `A` when `B` moves.

For a perpendicular pair, rotate the direction by opposite signed angles.

```
\coordinate (L) at ($(A)!7mm!90:(B)$);
\coordinate (R) at ($(A)!7mm!-90:(B)$);
```

This gives a local left and right that follow the orientation of segment `AB`.

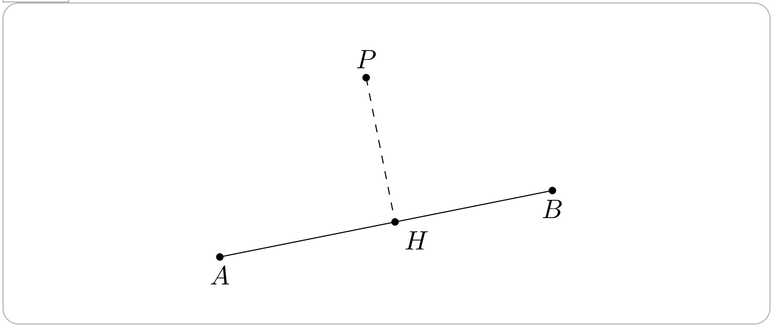
26.5 Projecting a Point onto a Line

The projection form is especially useful in Euclidean constructions.

```
\coordinate (H) at ($(A)!(P)!(B)$);
```

Here `H` is the perpendicular projection of `P` onto the line through `A` and `B`. The following picture makes the dependency visible.

See



Projection uses the infinite line, not only the segment. The foot may therefore lie beyond an endpoint.

26.6 Orthogonal Coordinates as Constraints

Mixed coordinates are more than shortcuts. They encode alignment constraints.

```
\coordinate (C) at (A -| B);
\coordinate (D) at (A |- B);
```

Use them to route a connector through a bend that follows both endpoints.

```
\draw[->] (A.east) -- (A.east -| B.north)
  -- (B.north);
```

If the nodes move, the bend remains orthogonal. A manually measured bend point would not preserve that relationship.

26.7 Naming and Reusing Intersections

Separate the construction paths from their visible styling.

```
\path[name path=p] (A)--(B);
\path[name path=q] (C)--(D);
\path[name intersections={of=p and q,by=I}];

\draw (A)--(B);
\draw (C)--(D);
\fill (I) circle (1.6pt);
```

This makes `I` an ordinary named coordinate. If the two paths have several intersections, the `by=` list can name them in the order TikZ finds them.

Common Mistake

TikZ finds intersections of the paths as specified. Two short segments may fail to meet even though their infinite supporting lines would intersect. Extend the construction paths when the mathematics requires full lines.

26.8 Worked Project: Altitudes of a Triangle

26.8.1 Stage 1: Define the Vertices and Feet

```
\coordinate (A) at (0,0);
\coordinate (B) at (5.4,.4);
\coordinate (C) at (1.7,3.6);
\coordinate (Ha) at ($(B)!(A)!(C)$);
\coordinate (Hb) at ($(A)!(B)!(C)$);
```

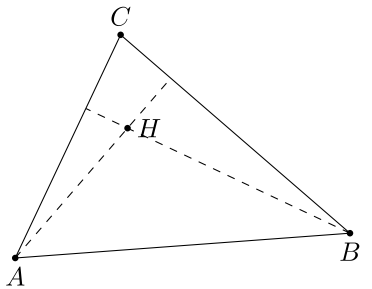
26.8.2 Stage 2: Name Two Altitudes

```
\path[name path=altitude A] (A)--(Ha);
\path[name path=altitude B] (B)--(Hb);
\path[name intersections={
  of=altitude A and altitude B,by=H
}];
```

26.8.3 The Complete Picture

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (5.4,.4);
  \coordinate (C) at (1.7,3.6);
  \coordinate (Ha) at ($(B)!(A)!(C)$);
  \coordinate (Hb) at ($(A)!(B)!(C)$);
  \path[name path=altitude A] (A)--(Ha);
  \path[name path=altitude B] (B)--(Hb);
  \path[name intersections={
    of=altitude A and altitude B,by=H
  }];
  \draw (A)--(B)--(C)--cycle;
  \draw[dashed] (A)--(Ha) (B)--(Hb);
  \fill (A) circle (1.5pt) node[below] {$A$};
  \fill (B) circle (1.5pt) node[below] {$B$};
  \fill (C) circle (1.5pt) node[above] {$C$};
  \fill (H) circle (1.5pt) node[right] {$H$};
\end{tikzpicture}
```

See



Move any vertex. The projection feet and orthocenter are recomputed from the new triangle.

26.9 Debugging Workshop

26.9.1 A Derived Point Does Not Move

Check whether it was entered numerically instead of calculated from named objects.

26.9.2 A Projection Falls Outside the Side

This can be mathematically correct for an obtuse triangle. Projection refers to the supporting line.

26.9.3 An Intersection Is Missing

Verify the path names, their scope, and whether the specified path portions actually cross.

26.9.4 The Orthogonal Bend Turns the Wrong Way

Exchange `-|` and `|-`; the two operators impose different coordinate combinations.

26.10 Exercises

1. Place a point 30 percent of the way from `A` to `B`.
2. Extrapolate a point beyond `B`.
3. Place two equal signed offsets on opposite sides of `A`.
4. Project a point onto a named line.

5. Construct an axis-aligned rectangle from two opposite corners.
6. Name the intersection of two construction paths.
7. Build one orthogonal connector from node anchors.
8. Revise a construction by moving one original point.

26.11 Solutions

26.11.1 Exercises 1–4

```
\coordinate (P) at ($(A)!.3!(B)$);
\coordinate (Q) at ($(A)!1.25!(B)$);
\coordinate (L) at ($(A)!6mm!90:(B)$);
\coordinate (R) at ($(A)!6mm!-90:(B)$);
\coordinate (H) at ($(A)!(P)!(B)$);
```

26.11.2 Exercises 5–7

```
\coordinate (C) at (A -| B);
\coordinate (D) at (A |- B);
\path[name path=p] (A)--(C);
\path[name path=q] (B)--(D);
\path[name intersections={of=p and q,by=I}];
\draw[->] (A.east) -| (B.north);
```

Exercise 8 changes an original coordinate and checks every derived object rather than repairing them independently.

26.12 ChatGPT Workshop

Ask ChatGPT to translate geometric statements into dependent coordinates.

1. Replace measured points by interpolation, distance, or projection forms.
2. Explain whether each value represents a fraction or a fixed distance.
3. Diagnose an intersection that disappears after a vertex moves.
4. Rewrite manual orthogonal bend points with mixed coordinates.
5. Propose a revision test that exposes hidden absolute coordinates.

26.13 Final Checklist

Before continuing to Chapter 27, check that you can

1. interpolate and extrapolate along a line;
2. distinguish fractions from fixed distances;
3. create signed perpendicular offsets;
4. project a point onto a line;
5. use mixed coordinates as alignment constraints;
6. name and reuse intersections; and
7. test a construction by moving its inputs.